

Paper for ENSIA Master's Graduation

Explainable AI-Powered Frameworks for OCT-Based Macular Disease Diagnosis : Public Benchmark Multimodal Analysis and Clinical Decision Support Strategy

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Abstract

Retinal diseases such as Age-Related Macular Degeneration (AMD), Diabetic Macular Edema (DME), and retinal vascular disorders are among the leading causes of irreversible visual impairment worldwide. Early diagnosis is essential for preserving vision, however, manual interpretation of Optical Coherence Tomography (OCT) examinations remains time-consuming, highly dependent on expert knowledge, and difficult to scale in routine clinical practice. Although recent deep learning approaches have shown promising performance in automated retinal disease analysis, many existing methods rely mainly on imaging data and often lack clinical interpretability and real-world validation. This work proposes two deep learning frameworks for OCT-based retinal disease diagnosis on both benchmark and real-world clinical data. The Framework I focuses on multimodal learning using the publicly available OCTDL dataset, where multiple visual encoders, metadata representations, and fusion strategies are evaluated. A Swin Transformer-Base combined with clinical metadata (age, sex, eye laterality) through dual cross-attention achieved the strongest performance, reaching a test accuracy of 93.50% and a weighted F1-score of 92.91%. Framework II addresses real-world clinical deployment using a private patient-level OCT dataset under weak supervision. A hierarchical two-stage framework based on Gated Attention Multiple Instance Learning (GA-MIL) was developed for patient-level diagnosis. Stage 1 performs binary screening to distinguish normal and pathological cases, achieving an accuracy of 96.88%, an F1-score of 98.21%. Stage 2 performs fine-grained disease classification for pathological cases and achieved a macro ROC-AUC of 79%. Explainability analyses using Grad-CAM, Integrated Gradients, and anatomically guided retinal layer segmentation further demonstrated that the proposed models consistently focus on clinically meaningful retinal structures. Overall, the results demonstrate that combining multimodal learning, weakly supervised patient-level modeling, and explainability improves the robustness and clinical applicability of OCT-based retinal disease diagnosis.

Keywords: *Optical Coherence Tomography, Macular Disease Diagnosis, Medical Image Analysis, Swin Transformer, Cross-Attention Fusion, Clinical Metadata Integration, Explainability, Grad-CAM, Integrated Gradients, Clinical Decision Support, patient-level diagnosis, Gated Attention, Multiple Instance Learning.*

1 Introduction

Retinal diseases are among the leading causes of visual impairment and irreversible blindness worldwide, affecting millions of individuals annually [1, 2]. Conditions such as Age-Related Macular Degeneration (AMD), Diabetic Macular Edema (DME), retinal vas-

cular occlusions, and other macular disorders progressively damage retinal structures and may lead to permanent vision loss if not detected early [3, 4]. The increasing prevalence of aging populations and systemic diseases such as diabetes has created a growing demand for accurate, scalable, and clinically reliable automated diagnostic systems.

Optical Coherence Tomography (OCT) has become a key imaging modality in ophthalmology because it provides non-invasive, high-resolution cross-sectional visualization of retinal anatomy [5–7]. OCT enables the identification of important pathological biomarkers such as retinal fluid accumulation, drusen deposits, and retinal layer disruptions that are essential for diagnosis and disease monitoring [2, 8]. However, manual interpretation remains time-consuming, requires expert knowledge, and is difficult to scale in routine clinical settings.

Recent advances in deep learning have significantly improved automated OCT analysis, with convolutional neural networks and transformer-based architectures demonstrating strong performance in learning retinal representations [9–11]. Nevertheless, many existing methods remain limited in clinical realism. Most approaches rely only on imaging information while overlooking complementary patient metadata, and are often evaluated using image-level annotations rather than patient-level clinical settings.

To address these limitations, this work proposes two frameworks for automated retinal disease diagnosis. The Framework I focuses on multimodal learning under benchmark conditions by integrating a transformer-based visual encoder with structured clinical metadata through bidirectional attention-based fusion for images classification on the OCTDL dataset. The Framework II targets real-world clinical scenarios using patient-level OCT examinations under weak supervision, where Multiple Instance Learning is employed to aggregate information using gated attention across OCT slices for reliable diagnosis.

To enhance transparency and support clinically reliable decision-making, the proposed framework integrates explainability mechanisms based on Grad-CAM and Integrated Gradients, and anatomically guided retinal layer interpretation using a DeepLabV3 model. This allows model predictions to be associated with clinically meaningful retinal structures.

Experimental results across benchmark and clinical settings demonstrate strong performance in terms of accuracy, robustness, calibration, and interpretability, highlighting the potential of the proposed methods for reliable AI-assisted retinal disease diagnosis in real-world environments.

2 Related Work

Optical Coherence Tomography (OCT) is one of the most widely used imaging modalities in ophthalmology because it provides high-resolution, cross-sectional visualization of retinal tissues in a non-invasive manner. By enabling micrometer-scale analysis of retinal layers, OCT facilitates the detection of structural abnormalities associated with diseases such as age-related macular degeneration (AMD), diabetic macular edema (DME), choroidal neovascularization (CNV), and glaucoma [5–7]. OCT is routinely used for diagnosis and disease monitoring by revealing biomarkers such

as retinal fluid accumulation, drusen deposits, retinal thinning, and layer disorganization. However, manual interpretation of OCT volumes remains challenging due to the large number of B-scans, inter-observer variability, and subtle pathological differences [1, 4].

Early automated OCT analysis relied on traditional image processing and handcrafted feature extraction methods, including retinal thickness measurements, texture descriptors, wavelet coefficients, and reflectivity features. These features were commonly classified using machine learning algorithms such as support vector machines, random forests, and k-nearest neighbors [7, 12]. Although these methods provided initial evidence for automated diagnosis, their performance depended heavily on segmentation quality and manual feature design.

The emergence of deep learning significantly transformed OCT analysis by enabling hierarchical feature extraction directly from image data. Convolutional Neural Networks (CNNs) became dominant for retinal disease classification. Kermany et al. demonstrated that an Inception-based model could classify OCT images with performance comparable to expert ophthalmologists [9]. Subsequent studies employed architectures such as VGG, ResNet, DenseNet, and EfficientNet, reporting strong classification performance and generalization capabilities [13, 14]. CNNs have also been extended to volumetric OCT analysis to model inter-slice relationships in complex retinal pathologies [15, 16].

Deep learning has also improved segmentation tasks in OCT imaging. U-Net and its variants became standard approaches due to their encoder-decoder structure and skip connections, allowing accurate localization of retinal layers and pathological regions [17, 18]. Later studies introduced multi-scale and attention-based architectures to improve robustness under challenging pathological conditions [19, 20].

More recently, transformer-based models have emerged as promising alternatives to CNNs by modeling long-range dependencies through self-attention mechanisms. Vision Transformers (ViT) and Swin Transformers have shown strong performance in retinal classification and segmentation tasks, particularly when pathological patterns span multiple retinal layers [10, 11]. Hybrid CNN-transformer architectures further combine local feature extraction with global contextual reasoning to improve diagnostic performance [21, 22].

Beyond image-only learning, recent studies have explored multimodal frameworks integrating OCT images with clinical information such as age, gender, visual acuity, and disease history. Existing methods commonly employ early, late, or intermediate fusion strategies [23, 24], while attention-based methods have been introduced to improve cross-modal interactions and diagnostic robustness [25].

Explainable artificial intelligence techniques have also become increasingly important for clinical adoption. Methods such as Grad-CAM, saliency maps, and attention visualization help identify retinal regions contributing to model predictions and improve trans-

parency in clinical decision-making [26, 27].

Research Gaps and Contributions

Despite significant progress in automated OCT analysis, several challenges remain. Most existing models rely exclusively on OCT images and do not incorporate patient-specific clinical information, despite its importance in real clinical decision-making [1, 2].

Although multimodal approaches have gained attention, many rely on simple fusion strategies such as feature concatenation or late fusion, which may not effectively model complex interactions between retinal imaging and clinical data [23, 24]. In addition, the use of transformer-based architectures with dual cross-attention mechanisms remains limited in multimodal OCT diagnosis [11, 25].

Another challenge concerns interpretability and clinical reliability. Existing explanation methods do not always guarantee clinically meaningful interpretation, and many models are trained using single-center datasets, limiting generalization across institutions and imaging devices [1, 26].

To address these limitations, this work proposes a multimodal retinal disease classification framework integrating OCT imaging with structured clinical metadata.

Attention-based fusion mechanisms are employed to model richer interactions between visual and clinical features, while explainability techniques are incorporated to improve transparency and support clinically meaningful decision-making. By combining multimodal learning, transformer-based feature extraction, and explainable AI, the proposed approach aims to improve diagnostic accuracy, robustness, and clinical applicability.

3 Methodology

This section presents the overall methodology of the proposed retinal disease diagnosis framework, evaluated through two complementary experimental settings on distinct datasets to distinguish controlled benchmarking from real-world clinical assessment. Framework I uses a public dataset as a controlled benchmark for systematic evaluation under standardized conditions, while Framework II relies on real-world clinical data to assess model performance and robustness under clinical variability, including differences in data quality, acquisition noise, and distributional shifts.

3.1 Framework I: Multimodal Learning Strategies

3.1.1 OCTDL Benchmark Description

To evaluate the proposed multimodal diagnostic framework under controlled benchmark conditions, experiments are conducted on the OCTDL dataset [28], a publicly available retinal OCT benchmark comprising

2,064 macular B-scan images collected from 821 patients. The dataset consists of seven retinal disease categories with a highly imbalanced class distribution:

Age-Related Macular Degeneration (AMD, 1,231 images), Normal (NO, 332 images), Epiretinal Membrane (ERM, 155 images), Diabetic Macular Edema (DME, 147 images), Retinal Vein Occlusion (RVO, 101 images), Vitreomacular Interface Disease (VID, 76 images), and Retinal Artery Occlusion (RAO, 22 images).

This imbalance reflects real-world clinical prevalence and introduces challenges for robust disease classification.

In addition to OCT images, the dataset provides structured patient-level metadata, including age, sex, and eye laterality, with annotations distinguishing left and right eye scans.

3.1.2 OCTDL Preprocessing Pipeline

A preprocessing pipeline is applied to standardize both imaging and clinical inputs prior to training. OCT images are denoised using a median filter to reduce speckle noise while preserving retinal structural boundaries. Each image is resized to 224×224 pixels and intensity-normalized to ensure consistent input distributions.

Clinical metadata is processed to handle missing or inconsistent entries. Continuous variables, such as age, are standardized using z-score normalization, while categorical variables, including sex and eye laterality, are encoded into numerical representations suitable for neural network processing. To prevent data leakage, a patient-wise splitting strategy is enforced, ensuring that all samples from a given patient belong to a single subset. The dataset is divided into training, validation, and test sets. Class imbalance is addressed using random oversampling applied only to the training set, combined with on-the-fly data augmentation to improve generalization.

3.1.3 Proposed Multimodal Framework

Based on preliminary comparative experiments, the Swin Transformer-Base (Swin-B) is selected as the visual encoder due to its strong capability in capturing both local retinal structures and long-range contextual dependencies. Its hierarchical shifted-window self-attention mechanism makes it particularly suitable for Optical Coherence Tomography (OCT) analysis, where diagnostically relevant patterns appear at multiple spatial scales.

In parallel, structured clinical metadata (age, sex, and eye laterality) is processed using a Multilayer Perceptron (MLP) encoder, which maps low-dimensional clinical variables into a compact latent representation aligned with the visual feature space.

Instead of simple feature concatenation, the multimodal fusion is performed using a calibrated Dual Cross-Attention mechanism. This module enables dual interaction between image and metadata embeddings,

allowing each modality to selectively attend to the most informative features of the other.

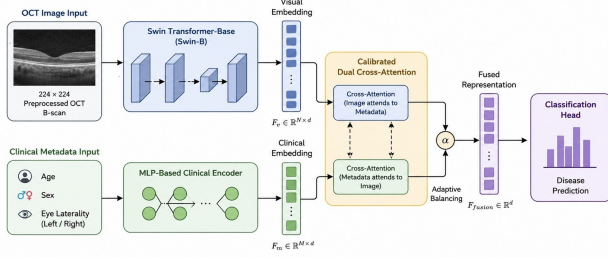


Figure 1: Overview of the proposed multimodal diagnostic framework. A Swin-B visual encoder and an MLP-based clinical encoder extract features from OCT images and structured metadata, which are fused using Dual Cross-Attention for final disease prediction.

Let $F_v \in \mathbb{R}^{N \times d}$ and $F_m \in \mathbb{R}^{M \times d}$ denote the visual and metadata embeddings. The cross-modal interactions are computed as:

$$CA_{v \leftarrow m} = \text{Softmax} \left(\frac{Q_v K_m^\top}{\sqrt{d}} \right) V_m \quad (1)$$

$$CA_{m \leftarrow v} = \text{Softmax} \left(\frac{Q_m K_v^\top}{\sqrt{d}} \right) V_v \quad (2)$$

where Q , K , and V are learned linear projections of the respective modality features.

The calibrated fusion is then obtained as:

$$F_{fusion} = \alpha CA_{v \leftarrow m} + (1 - \alpha) CA_{m \leftarrow v} \quad (3)$$

where $\alpha \in [0, 1]$ controls the contribution of each modality and stabilizes the fusion process.

The resulting fused representation is then forwarded to a fully connected classification head to perform final disease prediction. Figure 1 provides an overview of the complete architecture, including both encoding branches and the dual fusion mechanism.

3.1.4 Training Strategy

All models are initialized with ImageNet-pretrained weights and fine-tuned on the OCTDL dataset using transfer learning. Optimization is performed using the AdamW optimizer with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . A ReduceLROnPlateau scheduler is used for adaptive learning rate adjustment. Mixed-precision training is employed to improve efficiency, and early stopping is used to mitigate overfitting.

The model is trained using cross-entropy loss. Performance is evaluated on a patient-level independent test set using accuracy, weighted F1-score, sensitivity, and specificity.

3.1.5 Multimodal Explainability Analysis

To improve transparency and clinical interpretability of the proposed multimodal framework, an explainability module is integrated into the inference pipeline. Since

clinical adoption of artificial intelligence requires both high predictive performance and interpretable decision-making, multiple explainability techniques are employed to analyze visual retinal biomarkers and structured clinical feature contributions.

A. Visual Explainability Using Grad-CAM To identify retinal regions influencing model predictions, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the image branch of the multimodal model. Since the proposed framework employs a Swin Transformer as the visual backbone, the Grad-CAM implementation is adapted to transformer-based feature representations.

During inference, forward and backward hooks are attached to intermediate transformer layers to capture feature activations and gradient information. The resulting activation maps are projected onto the original OCT images to generate heatmaps highlighting the most discriminative retinal regions contributing to the decision.

B. Multimodal Explainability Using Integrated Gradients To further analyze the contribution of both imaging and structured clinical inputs, Integrated Gradients (IG) is applied to the full multimodal architecture. Attribution scores are computed separately for the OCT image branch and the metadata branch.

For image interpretation, pixel-level attribution maps highlight diagnostically relevant retinal structures. For metadata interpretation, feature-level attribution scores are computed for age, sex, and eye laterality, enabling quantitative assessment of the contribution of each clinical variable to the final prediction.

C. Anatomically Guided Explainability Through Retinal Layer Segmentation To further enhance clinical interpretability, an anatomically guided explainability framework is introduced by combining Grad-CAM activation maps with retinal layer segmentation. A DeepLabV3 model with a ResNet-50 backbone is used to segment major anatomical structures in each OCT scan.

The segmentation model identifies seven retinal layers: Internal Limiting Membrane (ILM), Retinal Nerve Fiber Layer (RNFL), Inner Plexiform Layer (IPL), Inner Nuclear Layer (INL), Outer Plexiform Layer (OPL), Outer Nuclear Layer (ONL), and Retinal Pigment Epithelium (RPE).

After segmentation, Grad-CAM heatmaps are spatially aligned with retinal layer masks. Layer-wise activation scores are computed by measuring the mean activation intensity within each anatomical region. This enables direct association between model attention and biologically meaningful retinal structures, providing stronger clinical interpretability.

By combining Grad-CAM, Integrated Gradients, and anatomically guided layer attribution, the proposed explainability pipeline provides spatial, multimodal, and anatomically grounded interpretability for

clinically reliable decision support.

3.2 Framework II: Clinical Decision Support Strategy

3.2.1 Real-World Clinical Dataset Description

The private retinal Optical Coherence Tomography (OCT) dataset collected in collaboration with the Ibn Al Haythem Ophthalmology Center is used to analyze real-world clinical conditions and guide the selection of appropriate modeling strategies for the diagnostic task. Unlike publicly curated benchmark datasets, this clinical dataset reflects routine ophthalmological practice and exhibits variability in acquisition protocols, patient demographics, image quality, and annotation completeness.

The dataset consists of anonymized macular OCT volumes acquired from patients presenting with both normal retinal anatomy and various pathological conditions. Each examination contains multiple B-scan slices extracted from volumetric OCT acquisitions, corresponding to different spatial locations across the macular region. Due to variations in acquisition settings and clinical protocols, the number of slices per patient is not constant, resulting in heterogeneous volumetric representations.

In contrast to the public benchmark, where each OCT B-scan is independently labeled, this clinical dataset follows a weakly supervised setting. Diagnostic annotations are provided at the patient or volume level, while slice-level labels are not available. This reflects real clinical practice, where diagnosis is typically based on the full OCT examination rather than individual slices.

The dataset supports a two-level diagnostic structure. The first level corresponds to binary screening, classifying each patient as either Normal or Abnormal. The second level involves fine-grained pathology identification for abnormal cases, where multiple retinal diseases may coexist within the same patient, leading to a multi-label diagnostic setting.

In addition to imaging data, partial clinical metadata are available for a subset of patients, including demographic variables such as age and sex. However, metadata completeness is inconsistent due to real-world clinical recording constraints.

Overall, this dataset is used to characterize real-world clinical variability and to inform the selection and adaptation of suitable modeling approaches for robust OCT-based diagnostic system design.

3.2.2 Real-World Clinical Dataset Preprocessing Pipeline

A dedicated preprocessing pipeline was developed to standardize raw OCT scans before model training. First, corrupted files, incomplete acquisitions, and non-OCT images were automatically removed based on decoding validation and metadata consistency checks.

To improve retinal structure visibility, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to enhance local contrast while preserving fine anatomical details. A region-of-interest extraction step was then performed to suppress irrelevant background regions and focus the model on diagnostically relevant retinal structures.

All OCT slices were resized to 384×384 pixels and normalized to the $[0, 1]$ intensity range before being passed to the feature extraction network.

Clinical metadata underwent expert-guided cleaning, including duplicate removal, correction of conflicting entries, and normalization of demographic variables. Continuous variables such as age were standardized using z-score normalization, while categorical variables such as sex were encoded numerically. Missing values were handled using explicit unknown categories when necessary.

To ensure fair evaluation and prevent data leakage, the dataset was partitioned using a strict patient-wise split into training, validation, and test subsets, ensuring that no patient appears across multiple splits.

3.2.3 Proposed Hierarchical Clinical Diagnostic Framework

To better reflect clinical decision-making, the proposed framework follows a hierarchical two-stage diagnostic strategy.

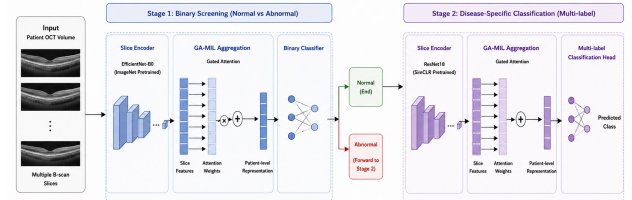


Figure 2: Overview of the proposed hierarchical clinical diagnostic framework. Stage 1 performs patient-level binary screening using EfficientNet-B0 with Gated Attention Multiple Instance Learning (GA-MIL), while abnormal cases are forwarded to Stage 2 for multi-label retinal disease classification using SimCLR-pretrained ResNet18 and GA-MIL.

Stage 1: Binary Screening.

The first stage of the proposed clinical framework performs early patient-level screening to distinguish between normal and pathological OCT examinations. Each patient examination is represented as an OCT volume composed of multiple B-scan slices acquired across the macular region. Since pathological biomarkers may only appear in a subset of slices, analyzing each image independently is insufficient for reliable diagnosis.

To address this challenge, each OCT slice is first processed independently using an EfficientNet

B0 backbone initialized with ImageNet pre-trained weights. This network extracts high-level visual representations that capture both local retinal microstructures and broader anatomical context.

The extracted slice-level feature vectors are then passed to a Gated Attention-Based Multiple Instance Learning (GA-MIL) module. Rather than assigning equal importance to all slices, the attention mechanism learns adaptive importance weights for each slice according to its diagnostic relevance. Slices containing pathological biomarkers such as retinal fluid, structural deformation, or layer disruption receive higher attention scores, while non-informative or peripheral slices receive lower weights.

A weighted aggregation of all slice embeddings is then performed to generate a single patient-level representation, which is finally passed to a binary classifier to predict whether the examination is normal or abnormal. This formulation allows the model to mimic clinical reasoning, where ophthalmologists focus on the most diagnostically relevant regions within the OCT volume.

Stage 2: Disease-Specific Classification.

Patients identified as pathological during Stage 1 are subsequently processed by a second diagnostic stage for pathology-specific classification. This stage is significantly more challenging, as different retinal diseases may share similar anatomical characteristics while also exhibiting subtle disease-specific biomarkers.

To improve feature learning under limited annotated clinical data, each OCT slice is encoded using a ResNet18 backbone pretrained through self-supervised contrastive learning using SimCLR. Unlike conventional supervised pretraining, SimCLR learns discriminative visual representations by maximizing agreement between augmented views of the same retinal image, allowing the encoder to capture robust retinal features without requiring dense manual annotations.

The resulting slice embeddings are then aggregated using the same GA-MIL mechanism. The attention module identifies the slices that contribute most strongly to disease discrimination and combines them into a patient-level feature representation.

Finally, this aggregated representation is passed to a disease classification head, producing patient level pathology predictions across multiple retinal disease categories. This two-stage design improves both screening efficiency and fine-grained diagnostic accuracy while remaining robust under weak supervision and real-world clinical variability.

3.2.4 Multiple Instance Learning Strategy

Since slice-level annotations are not available, each patient OCT volume is modeled as a bag of instances, where each instance corresponds to an individual B-scan slice. A Multiple Instance Learning (MIL) formulation is adopted, where the GA-MIL module assigns adaptive attention weights to each slice based on its diagnostic relevance.

The final patient-level representation is obtained through a weighted aggregation of slice embeddings, enabling the model to emphasize diagnostically informative slices while suppressing the contribution of non-informative or anatomically normal regions.

This formulation aligns with clinical practice, where pathological biomarkers may only be visible in a subset of slices within a macular OCT volume, while diagnosis is made at the patient level based on the complete examination.

3.2.5 Training Strategy

Both diagnostic stages were optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . Mixed-precision training was employed to improve computational efficiency.

To address class imbalance, weighted loss functions were incorporated during training. Early stopping based on validation performance was applied to prevent overfitting.

Final evaluation was performed at the patient level using Precision, Recall, F1-score, ROC-AUC, confusion matrix analysis, and calibration metrics in order to assess both predictive performance and clinical reliability.

4 Experiments and Results

4.1 Framework I: Experimental Multimodal Learning Models Evaluation

This framework evaluates the proposed multimodal diagnostic approach under controlled benchmark conditions using the OCTDL dataset. All experiments are conducted in a standardized setting to ensure fair and reproducible comparisons across different architectures.

Specifically, all models are trained using identical optimization hyperparameters and evaluated under the same patient-wise data split protocol, ensuring that no patient overlap occurs between training, validation, and test sets. This setup guarantees an unbiased assessment of model performance and enables a consistent comparison of different visual backbones and multimodal fusion strategies.

To evaluate the contribution of structured clinical information, the best-performing image encoder (Swin-B) is further analyzed in unimodal and multimodal settings. In the unimodal configuration, the model uses only OCT images, while in the multimodal configuration, image features are combined with patient metadata (age, sex, and eye laterality). The metadata is encoded using an MLP network, and fused with visual features using simple feature concatenation as a baseline.

Table 1 summarizes the performance comparison.

Table 1: Performance comparison between unimodal and multimodal learning using Swin-B.

Model	Accuracy	F1-score	Sensitivity	Specificity
Swin-B (OCT Only)	91.07	89.27	84.12	97.84
Swin-B + Metadata	93.30	91.03	86.36	98.40

The integration of clinical metadata led to a consistent improvement across all evaluation metrics. Compared with the unimodal model, the multimodal framework improved classification accuracy by approximately 2.2%, while also increasing sensitivity and specificity. These improvements indicate that patient-specific metadata provides complementary diagnostic information, particularly in visually ambiguous retinal conditions.

4.1.1 Comparison of Image Encoding Architectures

To assess the effectiveness of different image encoders, several state-of-the-art architectures are integrated with structured clinical metadata. The evaluated backbones include ResNet50, Vision Transformer (ViT), Swin Transformer-Tiny (Swin-T), and Swin Transformer-Base (Swin-B). Metadata features are encoded using a multilayer perceptron (MLP), and a baseline early-fusion strategy based on feature concatenation is adopted.

Table 2 presents the classification performance of the evaluated multimodal architectures.

Table 2: Performance comparison of multimodal architectures on the OCTDL dataset.

Model	Validation Accuracy	Test Accuracy
ResNet50 + Metadata	92.79	91.07
ViT + Metadata	90.38	88.34
Swin-T + Metadata	94.23	90.82
Swin-B + Metadata	94.71	93.30

The results indicate that transformer-based architectures generally outperform convolutional neural networks for OCT classification. ResNet50 achieves competitive performance, demonstrating effective retinal feature extraction, although its limited receptive field restricts global contextual modeling.

In contrast, the Vision Transformer (ViT) shows the lowest performance among the evaluated models, suggesting that the lack of inductive bias for local structures reduces its effectiveness on relatively small-scale medical datasets.

Swin Transformer variants achieve the strongest overall performance, highlighting the benefit of hierarchical self-attention in modeling retinal structures across multiple spatial scales. In particular, Swin-B achieves the best results, reaching a validation accuracy of 94.71% and a test accuracy of 93.30%.

Swin-T shows a larger generalization gap compared to Swin-B, indicating reduced stability under patient-wise splitting.

4.1.2 Multimodal Fusion Strategy Analysis

To study the integration of clinical metadata with retinal imaging features, several multimodal fusion strategies are evaluated using a fixed ResNet50 encoder and a consistent metadata representation, enabling an isolated assessment of the fusion mechanism. This analysis is conducted within a controlled ResNet50-based setting, whereas the final system adopts a Swin-B backbone as the visual encoder in the multimodal framework.

Table 3 summarizes the comparative performance of the evaluated fusion strategies across classification performance, calibration quality, and computational efficiency.

Table 3: Comparison of multimodal fusion strategies on OCTDL using ResNet50.

Metric	Concat	CrossAttn	M→I	I→M
F1 Macro	82.12	87.46	80.54	~60.00
F1 Weighted	93.08	90.99	91.61	~60.00
Precision	85.84	89.23	82.07	~60.00
Recall	81.25	86.74	81.66	~60.00
ROC-AUC	98.46	96.48	—	low
ECE (↓)	0.2600	0.0592	0.0812	high

The results show clear differences between fusion strategies across multiple evaluation dimensions. Simple concatenation achieves strong weighted F1-score and ROC-AUC; however, it suffers from poor calibration, as indicated by a high Expected Calibration Error (ECE = 0.2600), reflecting overconfident predictions.

Directional cross-attention variants improve performance and stability. The Metadata → Image configuration achieves moderate results with improved calibration, while the Image → Metadata direction exhibits severe performance degradation and unstable behavior, indicating ineffective cross-modal interaction in this direction.

Among all evaluated strategies, the proposed dual cross-attention mechanism achieves the best overall performance. It attains the highest Macro F1-score (0.8746), strong weighted F1-score (0.9099), and the lowest calibration error (ECE = 0.0592), while maintaining efficient inference time. These results demonstrate that bidirectional interaction between retinal imaging features and clinical metadata enables more robust and reliable multimodal representation learning.

4.1.3 Analysis of the Best Model

Based on the results of the ablation study, the proposed Swin-B based dual cross-attention model is selected as the final architecture for OCT classification.

The model achieves the best overall performance among all evaluated configurations, reaching a weighted F1-score of 0.9291. It also obtains strong precision (0.9342) and recall (0.9280), confirming its balanced performance across all classes.

In addition, the model demonstrates improved calibration compared to the ResNet50 baseline, with a

reduced Expected Calibration Error (ECE = 0.0291 after temperature scaling). This indicates more reliable confidence estimation in predictions.

Overall, these results confirm that the proposed Swin-B dual cross-attention model provides the most accurate, well-balanced, and clinically reliable performance for OCT-based disease classification.

4.1.4 Explainability Results

To assess the interpretability of the proposed diagnostic framework, qualitative and attribution-based analyses were performed on representative normal and pathological OCT samples. The evaluation includes visual saliency mapping, multimodal feature attribution, and anatomically guided retinal layer interpretation.

4.1.5 Visual Explainability Using Grad-CAM

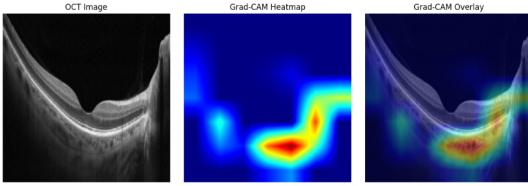


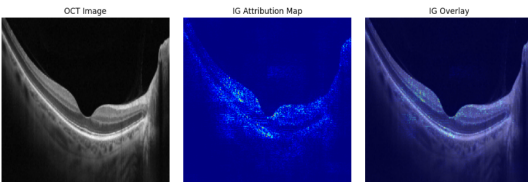
Figure 3: Grad-CAM visualization highlighting clinically relevant retinal regions.

Grad-CAM visualizations showed that the model consistently focused on clinically relevant retinal regions during decision-making. In pathological cases, high activation areas were primarily concentrated around the central macular region, highlighting structural abnormalities such as retinal layer disruptions, fluid accumulation, and localized deformations. In contrast, healthy cases exhibited more distributed activation patterns across anatomically regular retinal layers.

A qualitative comparison further showed that correctly classified samples were associated with focused and well-localized activation maps, whereas uncertain cases produced more diffuse attention patterns. These observations confirm that the model relies on clinically meaningful retinal biomarkers rather than irrelevant background features.

4.1.6 Multimodal Explainability Using Integrated Gradients

Integrated Gradients provided complementary feature-level interpretation across both imaging and clinical modalities. The attribution maps confirmed that OCT imaging remained the dominant source of diagnostic information, with strong positive contributions concentrated around pathological retinal regions.



```
{'prediction': 3,
 'class_name': 'NO',
 'confidence': 0.9179291725158691,
 'delta': -0.18816423416137695,
 'meta_attributions': {'age': 0.19625863432884216,
 'sex': -5.428928849226415e-05,
 'eye': 0.0}}
```

Figure 4: Integrated Gradients attribution maps. The upper panel shows visual attribution on the OCT image, while the lower panel presents the corresponding metadata feature contributions.

For the metadata branch, age consistently showed the highest attribution scores, indicating a meaningful contribution to the diagnostic process. In contrast, sex and eye laterality demonstrated lower attribution values across most samples, suggesting a more limited influence on final predictions.

4.1.7 Anatomically Guided Retinal Layer Attribution

To further validate the clinical relevance of model attention, Grad-CAM activation maps were combined with retinal layer segmentation generated using a DeepLabV3-ResNet50 segmentation model. The segmentation framework identified seven retinal layers, enabling anatomical localization of model attention.

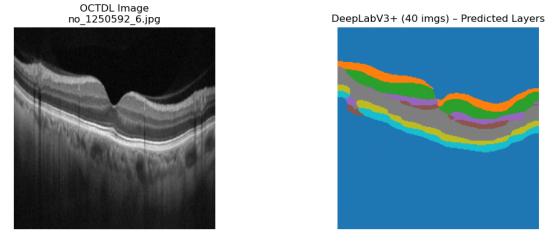


Figure 5: Retinal layer segmentation.

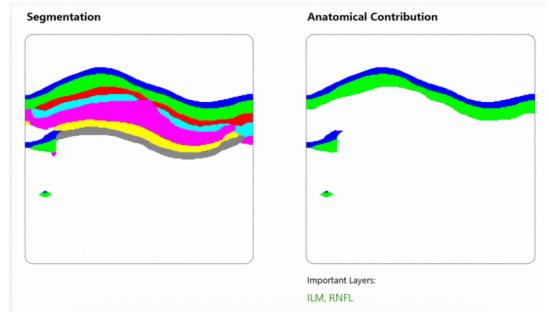


Figure 6: Anatomically guided Grad-CAM overlay showing the most important layers used by the model for prediction.

The overlay analysis revealed that the highest activation responses were consistently localized within clinically relevant retinal layers rather than background regions or imaging artifacts. In pathological cases, strong activation was frequently observed in inner and outer retinal layers associated with structural deformation and pathological fluid accumulation.

This anatomically guided analysis provides stronger clinical evidence that the proposed framework bases its predictions on biologically meaningful retinal microstructures.

4.1.8 Comparison with state of the art results

To further assess the effectiveness of the proposed multimodal framework, the final model was compared with representative state-of-the-art methods reported on the OCTDL benchmark. The comparison includes classical convolutional networks, transformer-based architectures, and recent multi-source training strategies.

Table 4: Comparison with published methods on the OCTDL benchmark.

Method	Architecture	Acc. (%)	W-F1	Setting
<i>CNN-based methods</i>				
Kulyabin et al. [29]	VGG16	88.10	0.852	OCTDL only
Kulyabin et al. [29]	ResNet50	88.20	0.857	OCTDL only
<i>Transformer-based methods</i>				
Hossain et al. [22]	SwinVFTR	92.10	0.912	OCTDL only
Li et al. [21]	Swin-Poly	93.80	–	External datasets
Zhang et al. [30]	VIT-Base	93.61	–	Multi-source training
<i>Proposed method</i>				
Ours	Swin-B + Metadata + Cross-Attention	93.50	0.9291	OCTDL + Metadata

The results confirm the superiority of transformer-based architectures over conventional CNN models for OCT classification. While recent transformer-based methods achieve strong performance, most remain limited to unimodal image-based learning or rely on external multi-source training. In contrast, the proposed framework achieves competitive performance using OCTDL data only while integrating structured clinical metadata through dual cross-attention. This multimodal design improves class balance, diagnostic robustness, and clinical relevance, establishing the proposed model as a strong and practical solution for real-world retinal disease classification.

4.2 Framework II: Two-Stage Clinical Decision Support Model Evaluation

Experiments are conducted on a private clinical OCT dataset collected at the Ibn Al Haytham Ophthalmology Center to evaluate the proposed framework under real-world conditions. The evaluation follows a two-stage diagnostic protocol, consisting of binary screening (normal vs. abnormal) followed by fine-grained disease-level analysis on abnormal cases.

4.2.1 Stage 1: Patient-Level Binary Screening

The first stage focuses on early screening by classifying each patient examination as either normal or pathological. During model development, three progressively refined MIL architectures were evaluated to improve screening robustness, feature representation, and clinical reliability.

Model A employed a ResNet18 backbone with a standard attention-based MIL aggregation mechanism. Although this baseline model achieved reasonable class separability, its sensitivity remained limited, indicating reduced capability in detecting abnormal patients.

Model B introduced improved attention weighting, stronger data augmentation, and better imbalance handling while maintaining the same ResNet18 backbone. This resulted in significant improvements in screening performance.

Finally, the proposed Model C replaced the backbone with EfficientNet-B0 and introduced a gated attention mechanism, mixed-precision training, weighted sampling, and AdamW optimization with cosine annealing. This final configuration achieved the strongest performance on the independent patient-level test set.

Table 5: Performance comparison of Stage 1 binary screening models.

Model	Accuracy	F1-score	ROC-AUC
Model A	62.00	73.00	92.00
Model B	84.00	90.00	97.70
Model C	96.88	98.21	98.75

A consistent improvement is observed across all evaluated architectures. The progression from Model A to Model C demonstrates the importance of stronger visual feature extraction, more discriminative attention-based aggregation, and improved optimization strategies in patient-level OCT screening.

To further assess the final screening model, additional clinical metrics were computed.

Table 6: Final screening performance of Model C.

Metric	Performance
Accuracy	96.88
F1-score	98.21
Sensitivity	96.49
Specificity	100.00
ROC-AUC	98.75
Brier Score	0.085
ECE	0.159

The final screening model achieved excellent clinical performance, combining very high sensitivity with perfect specificity. This is particularly important in screening applications, where missed pathological cases must be minimized while avoiding unnecessary false positive referrals. Calibration analysis further confirmed the reliability of the model’s probability estimates, making it suitable for clinical decision support.

To evaluate cross-domain generalization, the final Stage 1 model was further tested on the independent OCTDL dataset. Internal validation was performed using a held-out subset from the Ibn Al Haytham clinical dataset to assess performance under the same clinical distribution, while external validation on OCTDL was used to evaluate model robustness under domain shift.

Table 7: Internal and external validation performance of Stage 1 Model C.

Metric	Internal	External
Accuracy	96.88	87.27
F1-score	98.21	78.45
ROC-AUC	98.75	88.71
PR-AUC	–	86.11
Sensitivity	96.49	75.51
Specificity	100.00	92.47

Although a moderate performance drop is observed during external validation due to domain shift caused by differences in acquisition devices, patient populations, and annotation protocols, the model maintains strong discrimination and high specificity, confirming meaningful generalization beyond the development cohort. Nevertheless, several limitations remain. The framework is trained under weak supervision using only patient-level labels, meaning that attention maps provide valuable interpretability but should not be interpreted as precise lesion localization. In addition, while Model C demonstrates significantly improved calibration and confidence reliability, further post-hoc calibration strategies may further enhance its robustness for large-scale clinical deployment.

Analysis of the Best Model

Among all evaluated Stage 1 architectures, Model C achieved the strongest overall performance across classification accuracy, calibration, and external generalization. The combination of EfficientNet-B0, gated attention-based MIL, and optimized training strategies enabled more discriminative patient-level representations and improved robustness under real clinical conditions. These results establish Model C as the most reliable architecture for OCT-based binary screening and the selected backbone for subsequent clinical deployment.

4.2.2 Stage 2: Disease-Specific Classification

Patients identified as pathological during Stage 1 were subsequently processed by the second diagnostic stage for fine-grained disease classification. This stage is significantly more challenging due to multi-label pathology co-occurrence, overlapping anatomical biomarkers, and limited training samples.

To improve feature learning under limited annotated data, slice-level features were extracted using a ResNet18 encoder pretrained through self-supervised contrastive learning using SimCLR. The resulting slice embeddings were aggregated using the same GA-MIL mechanism to generate patient-level disease predictions across three pathological categories: Drusen, AMD, and Edema.

The final classification performance is summarized in Table 8.

Table 8: Stage 2 patient-level disease classification performance.

Metric	Drusen	AMD	Edema	Macro AUC
ROC-AUC	90	65	82	79

The model achieved the strongest performance on Drusen and Edema, indicating effective discrimination of visually distinctive retinal abnormalities. AMD remained the most challenging category, likely due to structural overlap with other pathological conditions and limited sample diversity. Despite these challenges,

the model maintained stable overall multi-label performance, demonstrating the feasibility of fine-grained patient-level disease classification under weak supervision.

The performance differences observed in Stage 2 are noticeably narrower compared with the binary screening results of Stage 1. This behavior reflects the increased complexity of fine-grained retinal disease classification under real clinical conditions. Unlike binary screening, Stage 2 requires the model to distinguish between multiple pathological conditions that often share overlapping anatomical biomarkers and may co-occur within the same patient examination. In addition, the disease-specific dataset contains fewer samples per class and exhibits stronger class imbalance, further increasing classification difficulty. Despite these challenges, the model maintains stable discriminative performance across all disease categories, indicating that the learned patient-level representations remain clinically meaningful even under weak supervision and limited annotated data.

4.2.3 Clinical Reliability and Generalization

Beyond predictive performance, the proposed patient-level framework demonstrated strong robustness under heterogeneous acquisition conditions and unseen patient populations. The gated attention mechanism enabled the model to automatically prioritize diagnostically relevant OCT slices while suppressing non-informative peripheral scans.

Overall, the experimental results demonstrate that the proposed hierarchical GA-MIL framework provides highly accurate screening, reliable pathology-specific classification, and meaningful generalization under realistic hospital conditions. These findings highlight the potential of the proposed system for real-world AI-assisted ophthalmic decision support.

5 Conclusion

In this work, we presented two complementary artificial intelligence frameworks for OCT-based retinal and macular disease diagnosis under benchmark and real-world clinical conditions.

The first contribution focused on multimodal diagnostic learning using a publicly available dataset. Through the evaluation of multiple image encoders, metadata representations, and fusion strategies, a Swin Transformer-Base visual encoder combined with structured clinical metadata and dual cross-attention achieved the strongest overall performance, demonstrating the effectiveness of multimodal learning for retinal disease classification.

The second and most clinically relevant contribution addressed real-world ophthalmic deployment using a private OCT dataset collected during routine medical practice. To address weak supervision, image variability, class imbalance, and patient-level heterogeneity, a hierarchical diagnostic pipeline based on Gated Atten-

tion Multiple Instance Learning was developed. Experimental results demonstrated strong screening performance, reliable calibration, and robust generalization across unseen cases, highlighting its potential for clinical decision support.

Explainability analyses using Grad-CAM, Integrated Gradients, and anatomically guided retinal layer segmentation further confirmed that the models consistently focused on clinically relevant retinal structures, improving interpretability and clinical trust.

Overall, this work demonstrates that combining multimodal learning, weakly supervised patient-level modeling, and explainable artificial intelligence can improve the robustness and clinical applicability of OCT-based retinal disease diagnosis.

Limitations: Despite the promising performance of the developed frameworks, several limitations remain. The real-world clinical dataset was collected from a single ophthalmology center, which may limit model generalization across institutions and imaging devices. In addition, the clinical framework relies on weak supervision where annotations are only available at the patient level, while slice-level labels remain unavailable. Metadata completeness was also inconsistent across patients, limiting richer multimodal interactions. Furthermore, some pathological categories contained relatively few samples, which may affect disease-specific learning performance.

Future Work: Future work will focus on validating the proposed methods using multi-center datasets from different hospitals and imaging systems to improve generalization. Longitudinal disease progression analysis may also support disease monitoring and prediction over time. Additional modalities such as OCTA, fundus imaging, and electronic clinical records may further enhance multimodal performance. Self-supervised and foundation models may also be explored to learn more robust retinal representations under limited annotation settings. Another promising direction involves leveraging publicly available OCT datasets with image-level labels to learn transferable slice-level representations that can later be adapted to clinical data, potentially improving slice prediction and reducing weak supervision limitations.

Data Availability

The publicly available OCT dataset used in this study is the Optical Coherence Tomography Dataset for Image-Based Deep Learning Methods (OCTDL), available through Mendeley Data at:

<https://data.mendeley.com/datasets/sncdhf53xc/4>

Additional details regarding the dataset description, acquisition protocol, and annotation process are provided in the original dataset publication by Kulyabin et al. [29].

The private clinical dataset used in this study is not publicly available due to institutional privacy regulations and patient confidentiality constraints.

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